

TIC 411978234

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 8.562175 d
Epoch (t0) 1661.7052 BTJD
Duration 0.0840 d
Depth 2.983 % (29,827 ppm)
SNR 132.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

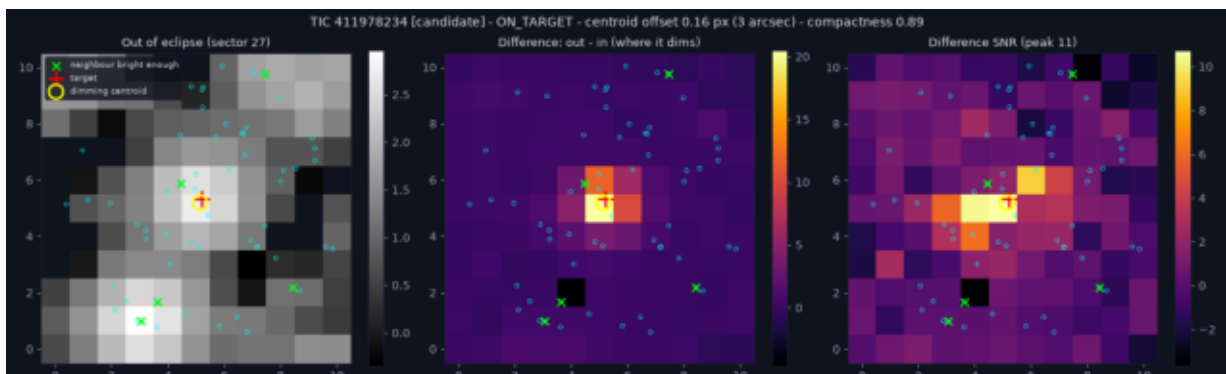
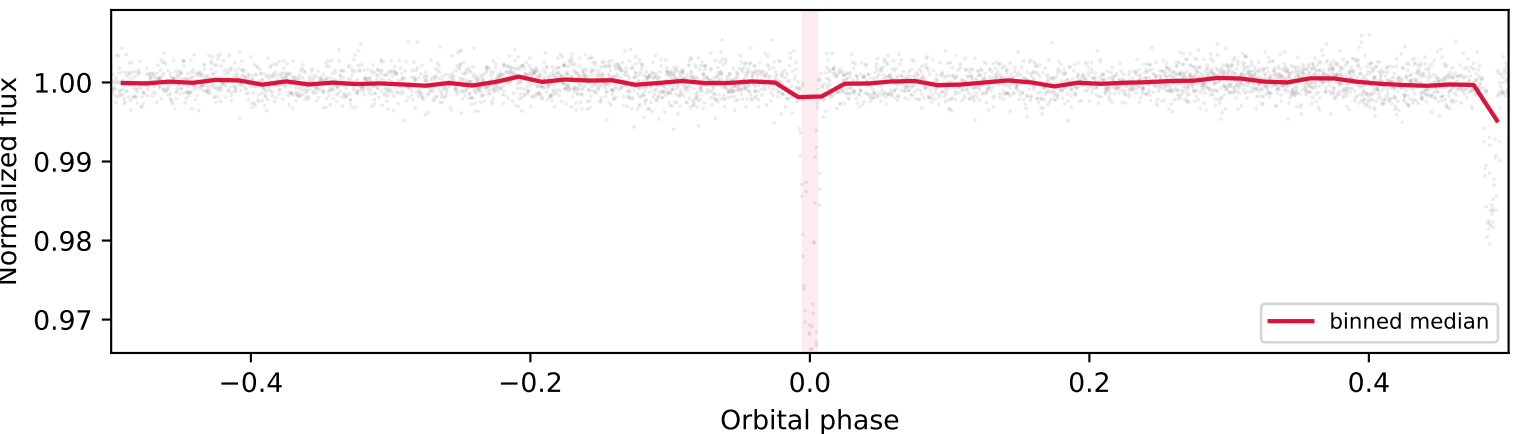
Eclipses observed 5 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 96.1 (gate ≥ 7.0)
Odd/even z-score 2.71 (consistent)
Secondary eclipse detected, phase 0.48, 9.6 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 27 light-curve dip in this sector: 73.2 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 10.74 (gate ≥ 5.0) Compactness 0.89 (gate ≥ 0.50)
Dimming centroid offset 0.16 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 411978230
0.96 px away, dTmag +3.23 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.